

KabadiAI: Project Overview, Progress and Winning Case

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This document explains what KabadiAI is, what has been built so far, what is planned next, and the honest case for why it can stand out. Status labels are literal: Done, Built but not deployed or tested live, Planned, Blocked.

1. The project in one paragraph

KabadiAI is a multilingual, voice-assisted mobile web app for kabadiwalas (informal scrap collectors and dealers). A kabadiwala photographs a lot of e-waste. The app lists the items and their condition, warns about hazards in plain language with a spoken alert, shows an indicative value range, checks whether a buyer's offer is low, fair or high, and shows the nearest authorized recyclers. A planned layer saves each real scan anonymously, so a dashboard and an analyst agent can show where e-waste and hazards cluster.

Pitch line: know what your lot is, what it is worth, and where to deliver it safely.

2. Why this problem

- India generates over 3 million tonnes of e-waste a year, and over 90% is processed by the informal sector (peer-reviewed case study on kabadiwalas, ScienceDirect, 2023).
- What is not recovered is often landfilled or burned. Open burning of boards and acid extraction of metal release toxic fumes.
- The E-Waste (Management) Rules, 2022 run on producer responsibility: producers buy EPR certificates, and only registered recyclers can generate them. One policy analysis notes the informal sector draws no recognition in the rules.
- Collectors mostly work in cash. One older study found only about 30% owned a smartphone. So the app is designed for voice and icons, and works for a collector with a phone or with someone's help.

We do not claim any subsidy, payment or official valuation. The app informs. It does not pay anyone.

3. Who it is for

- Primary user: the kabadiwala or scrap dealer. Wants to know what a lot is worth, what is dangerous, and where it can go legally.
- Secondary users (planned dashboard): authorized recyclers and city or NGO planners who want to see where e-waste and hazards concentrate.

4. What the app does today (the field app)

1. Choose a language (Hindi and English are complete).
2. Tap the big camera button and photograph the lot. The photo is resized in the browser before upload.
3. Optionally type the buyer's offer for the lot, using digits only.
4. See a result: items with icons, hazard warning in red with a speaker button, indicative value range, a traffic-light verdict on the offer, and the nearest recyclers with a directions link.
5. Share the summary on WhatsApp so a buyer can see it. The dealer needs no app.

5. What has been done

Infrastructure and backend (AWS SAM)

- Done: SAM project and deployed API (API Gateway, Lambda) with a scan endpoint. First deploy worked.
- Built, to confirm as deployed: DynamoDB tables for prices and recyclers, a seed script, price and recycler endpoints, and an agent endpoint.
- Built: agent logic rewritten after review. One vision call per photo. Prices, hazard text and recycler data come from tables and fixed text, never from the model. Rule-based hazard flags back up the model. Hazardous items are forced to zero value. A confidence gate asks for a better photo instead of guessing. An optional Strands agent mode orchestrates the same steps.
- Built, tested locally with stubbed tables, not deployed: the lot ledger. Each real analysis can be stored anonymously with location rounded to about 1 km, and a stats endpoint aggregates it. Mock lots are excluded from stats.
- Blocked: real photo analysis. The AWS account is still in new-account verification, so Bedrock calls fail. The app runs in a clearly labelled mock mode that returns fixed sample data.

Frontend (Next.js PWA)

- Built, not yet build-tested on the team's machine: a kabadiwala-first redesign with a high-contrast light theme for sunlight, large touch targets, icons on every item, a speaker button on key results, a help button, a language picker with native-script labels, and WhatsApp sharing with a fallback link.
- Voice currently uses the phone's own speech engine, so quality depends on installed voices.

Data and content

- Prices: indicative ranges per component with source notes. Several are placeholders and must be checked with a real dealer.
- Recyclers: five Delhi authorized recyclers from the DPCC and CPCB approved list (dated 2023), to be re-verified.
- Hazard messages: fixed text in English and Hindi. The Hindi is a draft that a native speaker must review.

Research and process

- Studied winning projects, including the tour's earlier winners and the 1st-place AWS AI Agent Global Hackathon winner, plus low-literacy interface research. The design choices below follow from that.
- Reviewed AI-generated code and fixed real bugs such as double-counted weights, duplicate vision calls, fragile result parsing, an API timeout trap and missing mock labels.

6. What we plan to do next

The QR handoff slip idea has been dropped. The ledger's receive route is not needed and can be removed or left unused.

Must do (P0)

1. Get Bedrock access working, then switch mock off and test with real photos. Without this, the demo and accuracy test cannot be real.
2. Deploy the lot ledger and confirm that real scans are saved.
3. Run the accuracy test: 40 or more real photos with hand-written ground truth. Report component accuracy, hazard recall, confidence behaviour and latency honestly.
4. Validate prices with one real scrap dealer, and watch him use the app if possible.
5. Native-speaker review of every safety string.
6. Record the 3-minute video and finish the writeup and README.

Should do (P1) 7. Insights dashboard: a map of where lots and hazards cluster, with totals by material. It reads the stats endpoint. 8. Analyst agent: a Strands agent that answers planning questions by chaining tools over the ledger (query, aggregate, chart or map). 9. Offline queue: capture a photo without internet, save it, and analyze it when the connection returns. The UI says honestly that analysis needs internet. 10. Design polish: Hindi font, vibration on hazards, items appearing one by one while scanning. 11. More languages: Punjabi, Bengali, Tamil, Telugu and Marathi, drafted with Amazon Translate and reviewed by native speakers. Safety text is always written and reviewed by people.

Could do (P2) 12. Amazon Location Service for an in-app map and road distance. 13. Polly voice for Hindi and Indian English.

Cut order if time runs short: analyst agent, then dashboard polish, then extra languages. Never cut the core scan, accurate hazards, the honest labelling, or the video.

7. Why this can stand out

It follows what actually wins.

- The 1st-place winner of the AWS AI Agent Global Hackathon was a waste app that closed a loop: photo reports, AI triage, staff tasks, hotspot maps and an agent that chains tools. It also worked offline and had real reports. KabadiAI follows the same pattern for e-waste: a field app, an accumulating ledger, a dashboard and an agent.
- The tour's own winners were agents built for one specific person, deployed live, with a strong safety-minded UI. KabadiAI is built for one person: the kabadiwala.

It is designed for its real user, with evidence. Research on low-literacy interfaces recommends graphical cues, voice annotation, local-language support, large controls and numeric-only input. The app follows these, and we plan to test it with a real kabadiwala.

It is honest where others overclaim.

- Safety text is fixed and reviewed, never generated by the model, and rule-based flags back up the model on hazards.
- Prices are labelled indicative, with source and date. Recycler coverage is labelled Delhi-NCR only.
- Mock data is always labelled and never used in the final demo or accuracy test.
- We report measured accuracy, including failures.

It makes real use of AWS. SAM, API Gateway, Lambda, DynamoDB, Bedrock, the Strands Agents SDK, and later Translate, Polly and Location Service, all visible in the demo.

It differentiates from existing pickup apps. Services such as Scrapdeal, ScrapUncle, Recykal and Karo Sambhav focus on pickup logistics. KabadiAI focuses on identification, hazard safety, price transparency and local-language voice guidance for the collector. This positioning must be verified before we claim it.

8. Mapping to the judging criteria

Criterion	Our case	Gap to close
Idea and impact	Specific user, documented problem, safety benefit	Talk to a real dealer and quote him
Built on AWS	Multiple services, SAM, Strands	Show them clearly in the video
Design and usability	Voice, icons, local language, sunlight-friendly	Test on a real phone and a real user
Execution	Working core flow in code	Real Bedrock output, deployed live, accuracy measured
Demo video	Clear story: photo, hazard voice, value, recyclers	Record on real analysis, within 3 minutes

9. Honest risks

- Bedrock access is still blocked. Everything real depends on it.
- The idea alone will not win. Execution quality, real data and a clean video will decide it.
- Weights from a photo are guesses, so values are rough. The UI says so.
- Recycler data is Delhi-NCR only and dated 2023.
- Language quality in safety text depends on native review that has not happened yet.
- Scope is growing. A working core beats an unfinished dashboard.
- Fewer than five percent of entrants reach any prize, and no one can promise a result. Prizes and the Amazon interview are separate decisions made by the panel.

10. Rules checklist

- All project code written after the hackathon opened. Practice material stays out of the repo.
- Repo is public with honest, frequent commits.
- AWS used and shown in the video.

- AI coding tools (Claude, Antigravity) listed in the writeup. Third-party material credited.
- Video under 3 minutes on YouTube, public or unlisted, tested signed out.
- Student verification complete for every team member.
- Submissions open Oct 11 at 8 AM. Confirm the deadline on the schedule page.

11. Immediate next steps

1. Resolve AWS account verification (email aws-verification@amazon.com and ask in the WeMakeDevs Discord).
2. Deploy the latest template and agent, then switch mock off once Bedrock works.
3. Start collecting real photos with GPS from different places, and book a dealer visit.
4. Find native speakers to review Hindi safety text and any further languages.